

On Relation between Graph and Topology

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Abstract

In this paper, we review the origin of graph theory and topology, think the semantic of link, and hence build a bridge between them. In addition, we find how ideas from graph theory evolve topology.

History

We know Euler solved *Königsberger Brückenproblem* opening graph theory, and found $V - E + F = 2$ opening topology. However, in his era they're under a historical branch called "analysis situs". This make us think if graph and topology are actually same thing.

Graph is Abstract

Since graph is more concrete, we first try endowing a topology on it. A naïve idea is just put the topology of shape on it, but same graph may represent different shapes! Thus we can't assume the meaning of graph alone. Like abstract set of simplices from points, we should "abstract" topology of graph from shape.

What is Link

Since links determine a graph, we think the semantic of "link": From intuition,

$$A \rightrightarrows B : A \text{ is arbitrary close to } B$$

Formulate to analysis' words,

$$A \rightrightarrows B : \forall \varepsilon > 0, \exists x, y \in A, B \text{ s.t. } d(x, y) < \varepsilon$$

Finally in topological saying,

$$A \rightrightarrows B : \forall N \in \mathcal{O}_B, N \cap A \neq \emptyset$$

Hence to make it consistent, the choice is:

- A is not Open : $A^c \rightrightarrows A$
- $'A$ is not Closed : $A \rightrightarrows A^c$

That is the common topology we want.

Foundation of Topology

We could see this topology is finite, but recall there is a natural basis

$$B_x := \bigcap \mathcal{O}_x$$

and bijection:

$$(x \rightrightarrows y : B_x \subseteq B_y) \rightarrow (B_x := \{y : y \rightrightarrows x\})$$

$$\text{PreOrder} \rightarrow \text{Finite Topology}$$

So we actually explained what is finite topology: In T_0 sense it's just graph! It also shows why some spaces are “bad”: Topology is thought studying “point”-set, but it may not! These make us rethink if some aspects in topology were misread.

Reference

- [1] Jerry Chen “On Subdivision and Meaning of Graph”
- [2] J. P. May “Finite Topological Spaces”